

SyncShift

Geolocation-aware attendance tracking for distributed and field-based workforces.

SyncShift is a production-grade clock-in/clock-out platform that validates employee location using server-side GPS checks, enforces configurable work schedules with grace periods, and provides administrators with rich reporting, audit trails, and CSV/PDF export capabilities.

Built as a monorepo with a **Node.js/Express** REST API, **React** web dashboard, and **React Native/Expo** mobile app — all backed by **PostgreSQL** and secured with RS256 JWTs, bcrypt hashing, and OS-level encrypted token storage.

Key Features

- 🌐 **Server-side GPS validation** — Haversine distance checks against configurable office geofences
- 🔒 **Secure dual-token auth** — Short-lived RS256 JWTs + opaque refresh tokens (httpOnly cookies on web, SecureStore on mobile)
- 📊 **Rich reporting** — Individual & team reports with preset date ranges and CSV/PDF export
- 🕒 **Schedule enforcement** — Shift times with configurable grace-minute tolerance and automatic late detection
- 📝 **Immutable audit trail** — All admin corrections logged with ID, timestamp, and mandatory justification
- 📱 **Cross-platform** — Web dashboard for admins, mobile-first experience for field employees

Tech Stack

Layer	Technology
Backend	Node.js · Express · PostgreSQL (raw SQL + knex)
Web	React · Axios · Context API
Mobile	React Native · Expo · SecureStore
Auth	JWT (RS256) · bcrypt · httpOnly cookies
Infra	Docker Compose · Nginx

Quick Start

```
# 1. Clone
git clone https://github.com/your-org/syncshift.git
cd syncshift

# 2. Configure environment
cp backend/.env.example backend/.env
# Edit with your DB credentials and JWT secrets

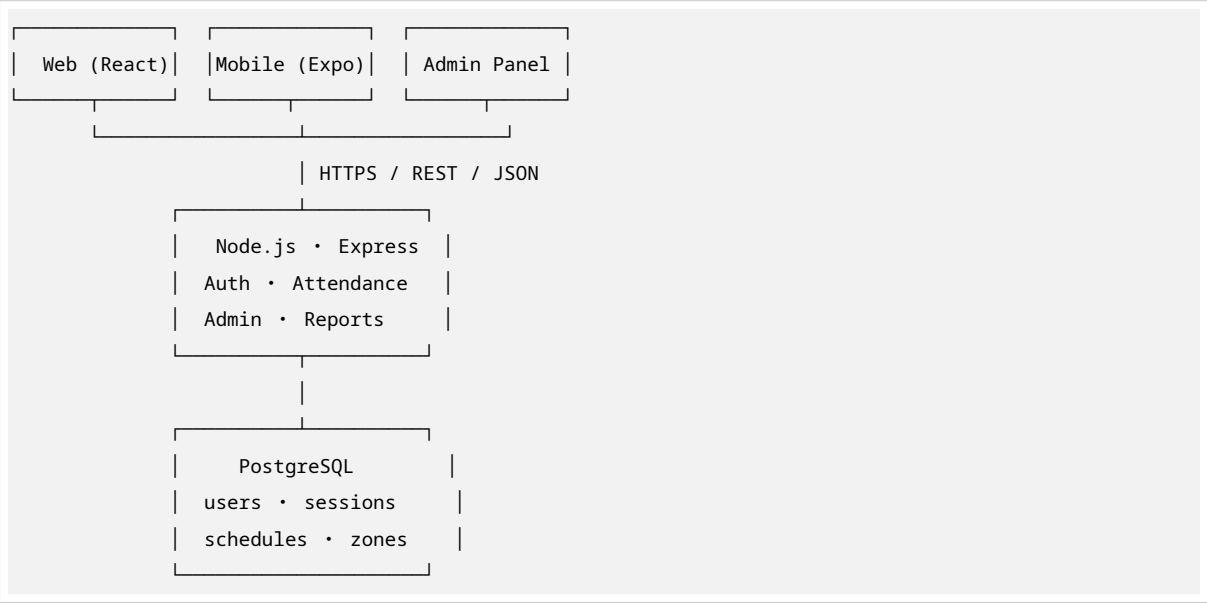
# 3. Start infrastructure
docker-compose -f infrastructure/docker-compose.yml up -d

# 4. Run migrations & seed
cd backend
npm run knex migrate:latest
```

```
../scripts/seed-admin.sh

# 5. Start services
cd backend && npm run dev      # API on :5000
cd frontend && npm start       # Web on :3000
cd mobile && npx expo start    # Mobile (Expo Go)
```

Architecture



Project Structure

```
syncshift/
├─ backend/      # API • services • repositories • middleware • jobs
├─ frontend/     # React web app • pages • components • hooks
├─ mobile/       # Expo app • screens • SecureStore • geolocation
├─ infrastructure/ # Docker • nginx • docker-compose
└─ scripts/      # setup • deploy • seed-admin
```

API Overview

Endpoint	Method	Access	Description
/api/auth/login	POST	Public	Access + refresh tokens
/api/attendance/signin	POST	Auth	Clock in with GPS validation
/api/attendance/signout	POST	Auth	Clock out •compute hours
/api/admin/reports/ user/:id	GET	Admin	Individual report + export
/api/admin/locations	CRUD	Admin	Office geofence management

See backend/ for full endpoint documentation.

Security

- **Passwords:** bcrypt, cost factor 12
- **Tokens:** RS256 JWTs (15m TTL) ·opaque refresh (7d TTL)
- **Web storage:** Access token in memory; refresh in httpOnly Secure SameSite=Strict cookie
- **Mobile storage:** Expo SecureStore (iOS Keychain / Android Keystore)
- **GPS:** Server-side Haversine validation — client coordinates never trusted alone
- **Roles:** Middleware-enforced per route; frontend state never trusted

Status

Phase: Design → Pre-Development

Version: 1.0

License: Confidential — internal use

Engineering Team ·2026